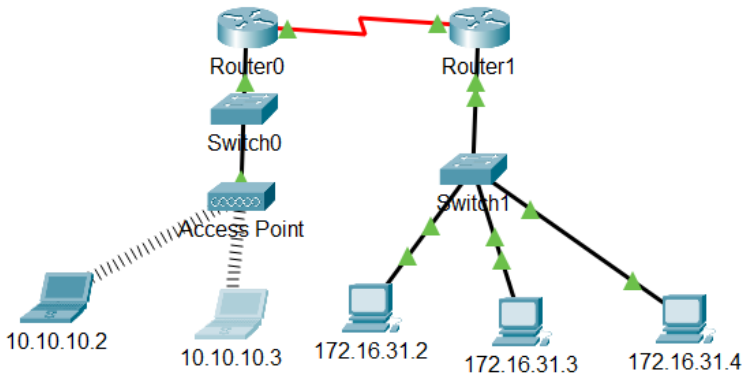
LAB 3: Packet Tracer - Examine Addressing Tables



Addressing Table

Device	Interface	MAC Address	Switch Interface
Router0	Gg0/0	0001.6458.2501	G0/1
	S0/0/0	N/A	N/A
Router1	G0/0	00E0.F7B1.8901	G0/1
	S0/0/0	N/A	N/A
10.10.10.2	Wireless	0060.2F84.4AB6	F0/2
10.10.10.3	Wireless	0060.4706.572B	F0/2
172.16.31.2	F0	000C.85CC.1DA7	F0/1
172.16.31.3	F0	0060.7036.2849	F0/2
172.16.31.4	G0	0002.1640.8D75	F0/3

Objectives

Part 1: Examine an ARP Request

Part 2: Examine a Switch MAC Address Table

Part 3: Examine the ARP Process in Remote Communications

Background

This activity is optimized for viewing PDUs. The devices are already configured. You will gather PDU information in simulation mode and answer a series of questions about the data you collect.

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Instructions

Part 1: Examine an ARP Request

Step 1: Generate ARP requests by pinging 172.16.31.3 from 172.16.31.2.

a. Click **172.16.31.2** and open the **Command Prompt**.

b. Enter the **arp -d** command to clear the ARP table.

c. Enter **Simulation** mode and enter the command **ping 172.16.31.3**. Two PDUs will be generated. The **ping** command cannot complete the ICMP packet without knowing the MAC address of the destination. So the computer sends an ARP broadcast frame to find the MAC address of the destination.

d. Click **Capture/Forward** once. The ARP PDU moves to **Switch1** while the ICMP PDU disappears, waiting for the ARP reply. Open the PDU and record the destination MAC address.

Is this address listed in the table above?
No it is not in the table above, FFFF.FFFF.FFFF address of local area network

e. Click **Capture/Forward** to move the PDU to the next device.

How many copies of the PDU did **Switch1** make?
it made three copies

What is the IP address of the device that accepted the PDU?
172.16.31.3

f. Open the PDU and examine Layer 2.

What happened to the source and destination MAC addresses?
the MAC address flipped and the broadcast address was overwritten with 31.3's unicast mac address.

g. Click **Capture/Forward** until the PDU returns to **172.16.31.2**.

How many copies of the PDU did the switch make during the ARP reply?

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only one copy

Step 2: Examine the ARP table.

a. Note that the ICMP packet reappears. Open the PDU and examine the MAC addresses.

Do the MAC addresses of the source and destination align with their IP addresses?
The src and MAC addresses correspond to the correct src and dst ip addresses base on the lab

b. Switch back to **Realtime** and the ping completes.

c. Click **172.16.31.2** and enter the **arp –a** command.

To what IP address does the MAC address entry correspond?
172.16.31.3 ip address corresponds to 0060.7036.2849 MAC

In general, when does an end device issue an ARP request?
when it does not know/have its MAC address in its MAC address table

Part 2: Examine a Switch MAC Address Table

Step 1: Generate additional traffic to populate the switch MAC address table.

a. From **172.16.31.2**, enter the ping **172.16.31.4** command.

b. Click **10.10.10.**2 and open the **Command Prompt**.

c. Enter the **ping 10.10.10.3** command.

How many replies were sent and received for each ping command?
4 sent and 4 received

Step 2: Examine the MAC address table on the switches.

a. Click **Switch1** and then the **CLI** tab. Enter the **show mac-address-table** command.

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Do the entries correspond to those in the table above?
the correspond a router and 31.3 and 31.4

b. Click **Switch0**, then the **CLI** tab. Enter the **show mac-address-table** command.

Do the entries correspond to those in the table above?
the addresses correspond to all end devices on the Router0 LAN

Why are two MAC addresses associated with the same port?
they are connected wirelessly to one access point which go to the same switch port

Part 3: Examine the ARP Process in Remote Communications

Step 1: Generate traffic to produce ARP traffic.

a. Click **172.16.31.2** and open the **Command Prompt**.

b. Enter the **ping 10.10.10.1** command.

c. Type **arp –a**.

What is the IP address of the new ARP table entry?
172.16.31.1

d. Enter **arp -d** to clear the ARP table and switch to **Simulation** mode.

e. Repeat the ping to 10.10.10.1.

How many PDUs appear?
two PDUs

f. Click **Capture/Forward**. Click the PDU that is now at **Switch1**.

What is the target destination IP destination address of the ARP request?
172.16.31.1

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g. The destination IP address is not 10.10.10.1.

Why?
Because we cleared the ARP table on 31.2. It needs to rebuild its MAC address of its local gateway.

Step 2: Examine the ARP table on Router1.

a. Switch to **Realtime** mode. Click **Router1** and then the **CLI** tab.

b. Enter privileged EXEC mode by entering the enable command:

Router> enable

Router#

c. Print the ARP table using the **show arp** command.

Is there an entry for **172.16.31.2**?
Yes

What happens to the first ping in a situation where the router responds to the ARP request?
the first ICMP packet times out and fails

Reflection Question

What are the differences between a switch and a router? Provide 3.
A router connects the entire home network to the external internet, whereas a switch only connects individual devices to each other inside the network. A router directs data between different networks using IP addresses, while a switch directs data between specific devices on the same network using MAC addresses. A router actively manages the network by providing security firewalls and assigning IP addresses, while a switch creates more wired ports for devices to plug into.

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